

XILIN GOL, China

WMO: 541020

Lat: 43.9501N

Lon: 116.1200E

Elev: 1004

StdP: 89.83

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.5	(c) -27.6	(d) -33.3	(e) 0.2	(f) -28.2	(g) -31.3	(h) 0.2	(i) -26.5	(j) 9.6	(k) -9.1	(l) 8.5	(m) -12.5	(n) 2.1	(o) 140	(p) 0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.3	(c) 32.8	(d) 17.1	(e) 30.9	(f) 16.6	(g) 29.1	(h) 16.0	(i) 20.0	(j) 27.3	(k) 19.1	(l) 26.2	(m) 18.2	(n) 25.2	(o) 4.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	14.5	22.7	16.8	13.6	22.1	15.8	12.7	21.5	62.4	27.2	59.0	26.2	56.0	25.2	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.9	(b) 8.6	(c) 7.6	(e) -32.1	(f) 35.6	(g) 2.7	(h) 2.0	(i) -34.0	(j) 37.1	(k) -35.6	(l) 38.2	(m) -37.1	(n) 39.4	(o) -39.0	(p) 40.8		
(a) 9.9	(b) 8.6	(c) 7.6	(e) -32.2	(f) 21.5	(g) 2.7	(h) 1.6	(i) -34.1	(j) 22.6	(k) -35.7	(l) 23.6	(m) -37.2	(n) 24.5	(o) -39.1	(p) 25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-18.1	-13.2	-3.9	6.0	13.3	19.2	22.3	20.2	14.0	4.7	-6.2	-15.2		
	DBStd	14.85	5.14	6.01	6.06	5.51	4.92	3.83	3.19	3.39	4.53	5.17	6.39	5.30		
	HDD10.0	3577	871	650	432	144	22	0	0	0	14	176	484	783		
	HDD18.3	5641	1129	883	690	372	169	35	4	19	141	423	734	1041		
	CDD10.0	1268	0	0	1	23	123	275	382	317	136	12	0	0		
	CDD18.3	289	0	0	0	1	13	60	127	77	12	0	0	0		
	CDH23.3	3056	0	0	0	22	228	668	1236	749	151	2	0	0		
	CDH26.7	1072	0	0	0	3	68	236	487	248	31	0	0	0		
(14) Wind	WSAvg	3.4	3.0	3.0	3.7	4.3	4.3	3.5	3.2	3.0	3.1	3.3	3.3	3.1		
(15) Precipitation	PrecAvg	277	2	2	5	7	23	42	83	68	28	11	4	2		
	PrecMax	480	9	9	14	30	95	98	282	158	67	68	18	7		
	PrecMin	155	0	0	0	0	1	12	10	6	1	0	0	0		
	PrecStd	67	2	2	4	7	21	20	49	39	19	12	4	2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.6	6.6	16.9	26.2	32.2	33.7	36.0	34.4	29.7	22.5	12.7	2.4		
		MCWB	-4.3	0.6	5.5	10.1	13.6	16.0	19.2	18.3	14.7	10.7	4.3	-1.9		
	2%	DB	-4.6	2.6	12.4	22.0	28.4	31.5	33.4	31.4	27.1	19.2	9.0	-2.0		
		MCWB	-6.9	-1.8	3.6	8.4	12.3	15.6	18.4	17.2	13.7	9.2	2.3	-4.9		
	5%	DB	-6.8	-0.2	9.1	19.1	25.8	29.5	31.4	29.3	24.9	16.4	6.2	-4.7		
		MCWB	-8.8	-3.8	1.6	7.3	11.3	14.9	17.8	16.5	13.0	7.6	0.9	-6.8		
	10%	DB	-9.5	-3.2	6.3	16.3	23.2	27.4	29.3	27.5	22.7	13.7	3.3	-7.0		
		MCWB	-10.9	-5.9	0.2	5.9	10.4	14.6	17.4	16.2	12.2	6.2	-0.8	-8.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.3	0.9	6.2	11.5	14.8	18.3	21.8	20.8	17.5	12.1	5.0	-1.8		
		MCDB	-1.1	5.5	14.2	24.3	25.7	26.5	28.1	29.3	25.3	20.2	12.0	2.2		
	2%	WB	-6.8	-1.5	3.9	9.1	13.5	17.4	20.5	19.7	15.4	9.9	2.7	-4.8		
		MCDB	-4.6	2.4	11.8	19.6	24.7	26.1	27.9	26.4	22.8	17.5	8.3	-2.2		
	5%	WB	-8.8	-3.8	2.0	7.8	12.4	16.6	19.6	18.7	14.2	8.3	1.0	-6.8		
		MCDB	-6.9	-0.3	8.9	17.6	23.2	25.3	27.0	25.0	22.2	15.1	5.9	-4.7		
	10%	WB	-11.0	-5.9	0.2	6.4	11.2	15.8	18.9	17.8	13.2	6.7	-0.7	-8.8		
		MCDB	-9.6	-3.2	6.0	15.7	21.3	23.8	26.2	24.3	20.8	13.2	3.3	-7.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	12.5	12.9	13.4	13.5	12.1	11.3	12.0	13.2	12.5	11.3	10.4		
		MCDBR	13.5	14.9	16.7	17.5	17.1	15.3	14.3	14.5	15.7	16.2	13.9	12.4		
		MCWBR	11.4	11.2	10.2	8.6	6.9	4.8	4.2	4.7	6.3	8.4	9.0	10.1		
		MCDBR	13.3	14.6	16.1	15.8	14.5	11.8	11.1	11.0	12.6	14.4	13.5	12.5		
	5% WB	MCWBR	11.3	11.2	10.0	8.3	6.3	4.3	4.1	4.4	5.7	8.3	9.0	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.268	0.303	0.350	0.405	0.468	0.491	0.450	0.435	0.384	0.348	0.295	0.276		
	taud		2.288	2.201	2.120	2.027	1.904	1.948	2.137	2.139	2.210	2.223	2.275	2.251		
	Ebn at Noon		859	880	877	855	808	790	820	818	833	813	810	803		
	Edh at Noon		82	109	136	162	189	182	149	144	123	106	83	78		
(44) All-Sky Solar Radiation	RadAvg		2.21	3.34	4.76	5.84	6.51	6.33	6.23	5.82	4.86	3.52	2.38	1.75		
	RadStd		0.18	0.23	0.32	0.37	0.41	0.43	0.24	0.20	0.26	0.25	0.17	0.19		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-1.89	N/A	N/A	-0.63	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	-1.64	N/A	N/A	-0.57	N/A	N/A	N/A	N/A

Nomenclature: See separate page